Christelinda Laureijs

Mount Allison University

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Education

MSc in Biology
Sackville, NB

Mount Allison University — *2026*

Thesis Title: Insulin in the rat dorsomedial hypothalamus decreases glutamate transmission but not feeding — Supervisor: Dr. Karen M. Crosby

BSc (Honours) in Biology, Minor in Biochemistry (4.3/4.3 CGPA)
Sackville, NB

Mount Allison University — *2024*

Thesis title: The effect of insulin on neuronal communication in the rat dorsomedial hypothalamus — Supervisor: Dr. Karen M. Crosby

Grants

ResearchNB Graduate Scholarship ($17,000)
Provincial: Research New Brunswick — *2024*

Canada Graduate Scholarship ($27,000)
National: Natural Sciences and Engineering Research Council of Canada (NSERC) — *2024*

ResearchNB Summer Studentship ($8,864)
Provincial: Research New Brunswick — *2023*

Independent Student Research Grant ($8,000)
Institutional: Mount Allison University — *2023*

Presentations

(Asterisks indicate presenter)

Summer Research Fair
Sackville, NB

Christelinda Laureijs*, Dr. Karen M. Crosby — *2026*

(Poster) How does insulin in the DMH affect neurons and behaviour?

Society for Neuroscience Annual Meeting
San Diego, CA

Dr. Karen M. Crosby*, Christelinda Laureijs — *2025*

(Poster) Insulin modulates glutamate transmission in the rat dorsomedial hypothalamus
(Poster) Enhancing learning and collaboration with a classroom magazine project

Atlantic Regional Comparative Physiology Workshop
St. Andrews, NB

Christelinda Laureijs*, Dr. Karen M. Crosby — *2025*

(Oral) Insulin in the brain: Exploring effects on neuronal activity and appetite

Summer Undergraduate Research Fair (SURF) for SHAD Canada
Sackville, NB

Christelinda Laureijs*, Dr. Karen M. Crosby — *2025*

(Oral) Exploring the effect of insulin injections into the dorsomedial hypothalamus on food intake and energy metabolism in rats

Canadian Association for Neuroscience Meeting
Toronto, ON

Christelinda Laureijs*, Dr. Karen M. Crosby — *2025*

(Poster) Efficiently plot and analyze electrophysiology data with patchclampplotteR

Canadian Association for Neuroscience Meeting
Toronto, ON

Dr. Karen M. Crosby*, Christelinda Laureijs — *2025*

(Poster) Insulin modulates glutamate transmission in the rat dorsomedial hypothalamus

Dalhousie Conference on University Teaching and Learning — *Halifax, NS*

Dr. Karen M. Crosby*, Christelinda Laureijs* — *2025*

(Oral) Enhancing learning & collaboration with a classroom magazine project

Spring into Teaching — *Sackville, NB*

Dr. Karen M. Crosby*, Christelinda Laureijs* — *2025*

(Oral) Enhancing learning & collaboration with a classroom magazine project

Open Science Skills in R — *Sackville, NB*

Christelinda Laureijs*, Elizabeth Stregger*, Dr. Julia Riley* — *2025*

(3-Part Workshop Series) Open Science Skills in R: Writing a Thesis in R

Research New Brunswick Health Research Symposium — *Moncton, NB*

Christelinda Laureijs*, Dr. Karen M. Crosby — *2024*

(Oral) Examining the effect of insulin on neuronal signalling in the rat dorsomedial hypothalamus

Atlantic Regional Comparative Physiology Workshop — *St. Andrews, NB*

Christelinda Laureijs*, Dr. Karen M. Crosby — *2024*

(Oral) Insulin decreases excitatory synaptic transmission in the rat dorsomedial hypothalamus – but how?

Summer Undergraduate Research Fair (SURF) for SHAD Canada — *Sackville, NB*

Christelinda Laureijs*, Dr. Karen M. Crosby — *2024*

(Oral) How will insulin injections in the rat brain affect food intake?

Gordon Research Conference: Hypothalamus — *Lewiston, ME*

Christelinda Laureijs*, Dr. Karen M. Crosby — *2024*

(Poster) Insulin modulates synaptic transmission in the rat dorsomedial hypothalamus

Obesity and Hypertension in Canada: From Science to Solutions — *Banff, AB*

Christelinda Laureijs, Dr. Karen M. Crosby* — *2024*

(Oral) Insulin Modulates Neuronal Signalling in the Rat Dorsomedial Hypothalamus

Biology and Applied Aquatic Sciences Joint Conference — *Halifax, NS*

Christelinda Laureijs*, Dr. Karen M. Crosby — *2024*

(Oral) Examining the relationship between insulin and neuronal activity in the rat dorsomedial hypothalamus

Research New Brunswick Health Research Week — *Saint John, NB*

Christelinda Laureijs*, Dr. Karen M. Crosby — *2023*

(Poster) The effect of insulin on neuronal communication in the rat dorsomedial hypothalamus

Summer Undergraduate Research Fair (SURF) for SHAD Canada — *Sackville, NB*

Christelinda Laureijs*, Dr. Karen M. Crosby — *2023*

(Oral) Does insulin affect neurons in the dorsomedial hypothalamus?

Society for Teaching and Learning in Higher Education (STLHE) Annual Conference — *Charlottetown, PEI*

Matthew Dunleavy*, Susan Andrews*, Carla VanBeselaere*, Christelinda Laureijs*, Shannon Goguen*, Denise Roy-Loar* — *2023*

(Oral) Together Time Story Sacks: Engaging Students-as-Partners to Build a Community-Based Intergenerational Literacy Program

Dalhousie Conference on University Teaching and Learning — *Halifax, NS*

Matthew Dunleavy*, Christelinda Laureijs*, Shannon Goguen*, Denise Roy-Loar* — *2023*

(Oral) Starting off Right: Students-as-Partners as the Foundation of a Community-Based Intergenerational Literacy Program

Innovations in Education Conference — *Hamilton, Ontario*

Dr. Jessica Riddell*, Dr. Toni Roberts*, Matthew Dunleavy*, Georges-Philippe Gadoury-Sansfaçon*, Shannon Heather Goguen*, Christelinda Laureijs*, Emma Trumble* — *2021*

(Oral) Students as Partners in Curricular and non-Curricular Work-Integrated Learning

Guest Lectures

Advanced Design & Statistical Analysis (BIOL 4711) *Mount Allison University*

Biology Department *Winter 2026*

Writing a Paper in R

Endocrinology (BIOL 4211) *Mount Allison University*

Biology Department *Winter 2026*

Insulin

Publications

Using a collaborative magazine project in undergraduate physiology classes [Manuscript in Preparation] *Advances in Physiology Education*

Karen M. Crosby, Christelinda Laureijs *2026*

A manuscript highlighting a teaching innovation in an upper year physiology course series.

Cholecystokinin and leptin act synergistically in the dorsomedial hypothalamus to affect glutamate signalling in male and female rats [Manuscript in review] *Journal of Neuroendocrinology*

Christelinda Laureijs, Emily Rushton, Kelsey MacKinnon, Karen M. Crosby *2026*

A primary research article using patch clamp electrophysiology to explore the effect of cholecystokinin and leptin on neurons in the dorsomedial hypothalamus

Insulin decreases glutamate signaling in the rat dorsomedial hypothalamus through insulin receptors and insulin-like growth factor 1 receptors (featured in APSselect) *Journal of Neurophysiology*

Christelinda Laureijs & Karen M. Crosby *2026*

A primary research article demonstrating the effect of insulin on glutamate transmission in the dorsomedial hypothalamus.

Reflecting on Vulnerability, Skill-Building and Identity in an Interdisciplinary SaP Project: Together Time Story Sacks *International Journal for Students as Partners*

Matthew Dunleavy, Susan Andrews, Carla VanBeselaere, Christelinda Laureijs, Shannon Goguen, Denise Roy-Loar *2024*

A case study reflecting on lessons learned from using the Students-as-Partners framework in a new literacy project called Together Time Story Sacks.

Using Embedded Simulations in Moodle Courses *Online Learning & Technology Consultants Program*

Christelinda Laureijs & Matthew Dunleavy *2021*

A tutorial (initially intended for faculty at Mount Allison University) demonstrating how to embed multimedia content into Moodle pages using custom HTML blocks.

OLTC Micro-Blogs – More than Just Moodle: Student-Educator Collaboration in the OLTC Program *Maple League of Universities*

Shannon Goguen & Christelinda Laureijs *2021*

Highlights the role of online learning technology consultants (OLTCs) in increasing meaningful student-educator partnerships across universities.

Packages

patchclampplotteR *GitHub*

Christelinda Laureijs *2024*

An R package developed to enable electrophysiologists to plot and analyze raw patch clamp data. https://christelinda-laureijs.github.io/patchclampplotteR/index.html

Employment

Research Assistant
Mount Allison University

Biology Department — *Summer 2026*

Performed whole-cell patch-clamp electrophysiology experiments and analyzed data for a collaborative project with researchers at the Université de Moncton.

Advanced Design & Statistical Analysis Graduate Student Teaching Assistant
Mount Allison University

Biology Department — *Winter 2026*

Helped students learn how to perform statistical analyses in R, troubleshoot coding errors, and plot and present their data.

Human Physiology Graduate Student Teaching Assistant
Mount Allison University

Biology Department — *Winter 2025 and Winter 2026*

Provided grades and detailed feedback to students on their written reports, posters, quizzes and oral presentations completed for the laboratory component of a challenging physiology course.

Animal Physiology Laboratory Graduate Student Teaching Assistant
Mount Allison University

Biology Department — *Fall 2024 & Fall 2025*

Helped upper-year Biology students collect consistent and reliable data in laboratory experiments by demonstrating the correct use of equipment, assisting students with technical issues, and training students in using R software for data analysis and data presentation.

Poster Designer
Mount Allison University

Modern Languages and Literatures Department — *2022-24*

Designed colourful and attractive posters for English-speaking students learning German. The posters include vocabulary posters (Animals, Food, and Home vocabulary) and posters highlighting unique idiomatic expressions in the German language. The posters are actively used in introductory German courses as testing material as well as to improve vocabulary retention and student engagement.

Notetaker
Mount Allison University

Meighen Centre — *Summer 2023 & Fall 2023*

Increased accessibility to course material by creating clear, well-organized notes for students through the Meighen Centre. Courses included Introduction to Material Cultures (Summer 2023) and Applied Genetics (Fall 2023).

Chemistry/Biochemistry/Biology Help Centre Teaching Assistant
Mount Allison University

Chemistry/Biology/Biochemistry Help Centre — *Fall 2022 - Winter 2023*

Helped students to succeed in first-year science courses through a twice-weekly help centre by coaching students as they discovered their own problem-solving approaches. Explained a wide variety of course concepts (in chemistry, biology and biochemistry), and demonstrated efficient ways to identify and resolve mistakes in practice question sets.

Research Assistant
Mount Allison University

Together Time Story Sacks — *2022-23*

Assisted with the launch of a pilot project assessing barriers to intergenerational literacy programs in rural New Brunswick. Compiled an annotated bibliography, summarized findings from the literature, analyzed interview transcripts, and shared major themes found in both the bibliography and interview transcripts. This work contributed to the widespread success of the project, with children and families across rural New Brunswick receiving Story Sacks and access to literacy materials.

Biology Laboratory Teaching Assistant
Mount Allison University

Biology Department — *Fall 2022*

Assisted students during the laboratory portion of a first-year introductory biology course (BIOL 1001) by demonstrating safe laboratory techniques, and helping students to successfully complete experiments. Answered student questions, provided guidance for assignments, and encouraged students to draw connections between course content and lab activities.

Piano & Music Theory Teacher
Amherst, NS

Self-Employed — *Summer 2022, 2024*

Taught private piano lessons in a one-on-one setting. Created lesson plans, corrected music theory assignments, and composed pop song arrangements at a level appropriate for the student's current abilities.

Biochemistry Teaching Assistant
Mount Allison University

Biochemistry Department — *Fall 2021*

Managed a drop-in help centre for students taking an Introductory Biochemistry course (BIOC 1001) at Mount Allison University. Explained course concepts, clarified misunderstandings, and demonstrated to students how they could approach problem solving on their own. Answered questions on Moodle (an online learning platform) and created illustrations and diagrams to assist with key concepts.

Online Learning Technology Consultant (OLTC)

Mount Allison University

Maple League of Universities

Summer 2021 - Fall 2021

Supported faculty as they adapted to online learning during the pandemic, by working with them to address their teaching goals and technical issues. Designed course Moodle pages, created quizzes and other online assessment tools and taught faculty how to use a wide variety of online learning and teaching tools. Performed syllabus audits and discussed best practices for learning and accessibility.

Academic Leadership

Animal Care Committee

Mount Allison University

Department of Biology

2023-Current

Served as the undergraduate (2023-24) and graduate (2024-26) representative on the animal care committee. Reviewed project proposals and renewals, participated in facility inspections and joined in discussions related to best practices for animal use on campus.

Lab Training

Mount Allison University

Crosby Lab

2023-Current

Trained Honours student Emily Rushton (2024-25) and Independent Studies student Jasmine Pannozzo-Oddy (2023-24) in conducting whole-cell patch-clamp electrophysiology experiments and collecting and analyzing data, which resulted in conference presentations and a publication (manuscript in preparation). Assisted lab members with troubleshooting equipment malfunctions, analyzing and plotting data, and preparing for presentations. Also trained two undergraduates (Kate Weldon & Selena Winchester) in lab practices such as making solutions, preparing the lab for experiments, and maintaining lab cleanliness.

Grad Studies Committee

Mount Allison University

Graduate Studies

2024-Current

Served as the graduate student representative on the Grad Studies Committee, assisting with sharing student perspectives and resolving administrative issues relating to the graduate studies program.

Graduate Student Association

Mount Allison University

Graduate Studies

2024-2026

Served as the Co-President of the Graduate Student Association, where I collaborated with fellow graduate students and Mount Allison faculty and staff to organize events during the year that foster a sense of community among graduate students.

Volunteering

Together Time Story Sacks Volunteer

Mount Allison University

Together Time Story Sacks

Fall 2024

Assisted with creating and managing science stations for preschool children visiting Mount Allison University for a day of literacy-focused activities. These stations allowed children to learn basic science concepts, interact with interesting materials from nature, and watch demonstrations of colourful scientific experiments.

Poster Designer

Mount Allison University

Modern Languages and Literatures Department

2022

Designed colourful and attractive vocabulary posters about Animals and Food for English-speaking students learning German. The posters are actively used in introductory German courses as testing material as well as to improve vocabulary retention and student engagement.

Neuroscience Research Collaborator

Mount Allison University

Université de Moncton

2024 and 2026

Collaborated with researchers at the Université de Moncton on a neuroscience research project. Used my experience and technical skills to perform whole-cell patch clamp electrophysiology, collect and image tissue samples, and analyze, present, and share data with the research team.

Recruiter

Mount Allison University

Canada-Wide Science Fair

Summer 2025

Created a fun and engaging display table and booth highlighting the science programs available at Mount Allison University, and presented this during the Canada-Wide Science Fair, which is the largest youth science fair in Canada.

Campus Welcome Person

Mount Allison University

Science Departments *2022*

Participated in welcome events throughout the year. This includes participating at the Biology display table, welcoming and assisting families with directions at Open House, and leading first-year students on Science Treks to introduce them to campus labs, instructors and faculty, and being available to meet with prospective honours students and discuss research opportunities.

Crosby Lab Mentor

Mount Allison University

Crosby Lab *2023-2026*

Answered questions from students, provided technical support in the lab, assisted with making solutions and performing lab tasks, helped students resolve coding issues, provided feedback on student posters, and demonstrated presentation and coding skills for students.

Awards

Academic Distinction

(Teaching Assistant Award) Outstanding Biology TA Award *2025*

Awarded to a teaching assistant that goes above and beyond expectations, actively participates in labs to help their peers understand material, and makes labs more fun.

(Academic Award) Governor General's Silver Medal *2024*

Awarded to the undergraduate student in the graduating class who has the highest average of all those receiving a first degree.

(Academic Award) Alumni Academic Achievement Award *2024*

Awarded to the graduate with the most distinguished academic record.

(Academic Award) Pravin Varma Gold Medal in Science ($500) *2024*

Awarded to the student graduating with the highest record in all subjects counting for the Bachelor of Science Degree with Honours.

(Academic Award) Harold Melville Rogers Memorial Gold Medal *2024*

Awarded to students of marked brilliance in Biology who possess those qualities and ideals that mark the highest type of university student and scientist.

(Academic Award) Biology Prize for Research Excellence ($250) *2024*

Awarded to Biology students whose academic standing is high and who show the promise of the ability to do original research.

(Oral Presentation Award) The David S. Fensom Prize (1st place; $500) *2024*

Awarded to students who have given the best public presentations of their Biology 4950 or Biology 4990 Honours work during the thesis presentations.

(Writing Award) Nova Scotian Institute of Science Student Science and Technology Communication Competition ($800) *2024*

Awarded to a Nova Scotian student who wrote the highest scoring research paper on an original topic. The papers are judged according to scientific merit, effective communication skills, and accessibility to a non-specialist audience.

(Oral Presentation Award) Science Atlantic Undergraduate Research Award for Oral Presentation ($200) *2024*

Awarded to the student (or students) giving the best research presentation(s) at an annual Science Atlantic conference.

Dean's List (x 4) *2021-23*

Applied to full-time students who receive at least a 3.7 SGPA.

(Academic Award) The Harold Chapman Steeves and Annalea Steeves Memorial Prize ($1,094) *2023*

Awarded to an outstanding student majoring or honouring in biology.

(Academic Award) William Thomas Ross Flemington Scholarship ($2,850) *2022*

Awarded to a student entering third year with a good academic record and interest & skill in lab work.

(Academic Award) Weeks Prize ($50) *2021*

Awarded to a student with the highest academic standing in Biology 1001 & 1501.

Scholarships

(Writing Award) IWK Future Health and Research Innovation Scholarship ($2,500) *2020*

Awarded to students in the Maritime provinces based on a research paper about a healthcare topic and its impact on families in the Maritimes.

Mount Allison Entrance Scholarship ($16,000) *2020*

Awarded to incoming students based on academic performance, leadership, extracurricular involvement, community service & work experience.

(Writing Award) Credit Counselling Services of Atlantic Canada Scholarship ($1,000) *2020*

Awarded to students based on an essay about responsible financial management.

(Film Award) Sable Island Scholarship ($1,500) *2020*

Awarded to a grade 12 student in Nova Scotia based on a creative art submission highlighting a breaking news story about Sable Island. I created a newsreel-style video titled "Connections to Sable Island" about Sable Island's role during World War II.

Showcase Productions Bursary ($500) *2020*

Awarded to a student based on volunteer contributions to the Showcase Productions society and a written essay on future goals.

Piano Competitions

(Piano Performance Award) 3rd Place | Intermediate Piano Competition at the New Brunswick Music Federation of Music Festivals *2019*

Awarded to a performer who placed third in the provincial competition in the intermediate piano class (ages 14-18).

(Piano Performance Award) 2nd Place | Junior Piano Competition at the Nova Scotia Music Federation Provincials *2019*

Awarded to a performer who placed second in the provincial competition in the junior piano class of the Nova Scotia Provincials.

(Piano Performance Award) Morrison Rose Bowl for Outstanding Senior Performance *2019*

Awarded for outstanding performances in the senior piano division of the Truro Music Festival. Festival. Students are also recommended to compete in the junior class of the Nova Scotia Provincials.

(Piano Performance Award) 2nd Place | Chamber Class Intermediate Competition at the New Brunswick Music Federation of Music Festivals *2017*

Awarded to a chamber group who placed second in the provincial competition in the intermediate chamber class.

(Piano Performance Award) 1st Place | Canadian Composer Intermediate Competition (x4) *2016-19*

Awarded to a performer who placed first in the provincial competition in the intermediate Canadian composer piano class (ages 14-18).

(Piano Performance Award) 2nd place | Junior Piano Competition at the New Brunswick Music Federation of Music Festivals *2014, 2019*

Awarded to a performer who placed second in the provincial competition in the junior piano class (< 14 years old).

(Piano Performance Award) Sackville Music Festival Award for Senior Piano Sight Reading *2018*

Awarded to first-place finalists with a grade above 85 in the senior piano sight reading competition.

(Piano Performance Award) Sackville Music Festival Awards (x7) *2013-19*

Awarded to first-place finalists in Sackville Music Festival. Recipients are also nominated to perform at the New Brunswick Music Federation Provincials.

(Piano Performance Award) Mount Allison Local Centre Piano Examinations Celebration of Talent Award (Grades 8-10) *2014-17*

Awarded to the student who received the highest mark across the province in their piano level in the Local Centre Piano Examinations.

Art, Poetry, and Film Contests

(Film Award) Not Your Grandfather's Mining Best 30-Second Commercial ($750) *2020*

Awarded based on a 30-second commercial about the next steps required for maintaining an environmentally responsible mining industry in Nova Scotia.

(Film Award) Not Your Grandfather's Mining Best 30-Second Commercial ($750) *2018*

Awarded based on a 30-second commercial educating Nova Scotians about the mining industry.

(Writing Award) First Place in Get to Know Your Wild Neighbors Poetry Contest *2019*

Awarded to the top entry in a poetry contest designed to encourage exploration in nature.

(Photography Award) Grand Prize winner of Get to Know Your Wild Neighbors Photography Contest *2018*

Awarded to the top entry in a photography contest designed to increase awareness about the wonder of nature.

(Film Award) Nourish Nova Scotia Food and Film Challenge Winner ($500 camera package) *2017*

Awarded based on a film about balancing food security with sustainability.